# Anantha Lakshmi Institute of Technology & Sciences

(AUTONOMOUS)

Near S.K. University, Itikalapalli, Ananthapuramu - 515721

## I B.Tech II Semester I-MID Examinations, May-2025

Branch: COMPUTER SCIENCE AND ENGINEERING | Max. Marks: 40

Subject: DIFFERENTIAL EQUATIONS & VECTOR CALCULUS (24A1BTH52021) | Time: 90 Minutes

### PART - A (Answer All Subquestions) | Total: 10 Marks

| Q.No. | Question Description | Marks |
|---|---|---|
| 1(a) | Form the partial differential equation by eliminating arbitrary functions from z = f(x^2 + y^2). | 2 Marks |
| 1(b) | Solve the linear differential equation (D^2 + 4D + 4)y = 0 with constant coefficients. | 2 Marks |
| 1(c) | Find a unit normal vector to the scalar surface xy + yz + zx = 3 at the point (1, 1, 1). | 2 Marks |
| 1(d) | If F = 2xz^2 i - yz j + 3xz k, find curl F at the point (1, 1, 1). | 2 Marks |
| 1(e) | State Green's theorem in a plane relating line integral and double surface integral. | 2 Marks |

### PART - B (Answer ONE Question from each Pair) | 3 × 10 = 30 Marks

| Q.No. | Main Descriptive Questions | Marks |
|---|---|---|
| 2. | State and prove the existence and uniqueness theorem for first-order linear differential equations and find the general solution of (x^2 + 1) dy/dx + 2xy = 4x^2. | 10 Marks |
| | [ OR ] | |
| 3. | Solve the Lagrange linear partial differential equation (x^2 - y^2 - z^2)p + 2xy q = 2xz and determine the integral surfaces. | 10 Marks |
| 4. | Find the directional derivative of f(x, y, z) = 2xy + z^2 at the point P(1, -1, 3) in the direction of the vector PQ where Q is (5, 0, 4). | 10 Marks |
| | [ OR ] | |
| 5. | Prove that the vector field F = (y^2 cos x + z^3)i + (2y sin x - 4)j + (3xz^2 + 2)k is conservative and find its scalar potential phi. | 10 Marks |
| 6. | Verify Green's theorem in the plane for integral (3x^2 - 8y^2)dx + (4y - 6xy)dy where C is the boundary of the region bounded by x=0, y=0 and x+y=1. | 10 Marks |
| | [ OR ] | |
| 7. | Using Stokes' theorem, evaluate the line integral of F = (x + y)i + (2x - z)j + (y + z)k along the triangle with vertices (2, 0, 0), (0, 3, 0) and (0, 0, 6). | 10 Marks |

*** End of Question Paper ***